



Started on	Wednesday, 5 June 2024, 2:31 AM
State	Finished
Completed on	Wednesday, 5 June 2024, 3:50 AM
Time taken	1 hour 19 mins
Marks	61.00/101.00
Grade	60.40 out of 100.00

QUESTION 1

Correct

Mark 1.00 out of 1.00

Assume that p, x, y, and z are int variables with x = 1, y = 10, z = -3. Which of the following is true after the following statement is executed? `p = (x++ * y-- ) + ++z;`

- ☐ a. p = 17
- ☐ b. C. p = 22
- ☒ c. D. p = 8 ✓
- ☐ d. p = 16

The correct answer is: D. p = 8

QUESTION 2

Incorrect

Mark 0.00 out of 1.00

What will happen in the following C++ code snippet? `int a = 100, b = 200; int *p = &a, *q = &b; p = q;`

- ☐ a. q now points to a
- ☐ b. a is assigned to b
- ☐ c. p now points to b
- ☒ d. b is assigned to a ✗

The correct answer is: p now points to b

QUESTION 3

Correct

Mark 1.00 out of 1.00

Which of the following best describes the statement "C++ is a case sensitive programming language"?

- ☐ a. Every statement in C++ ends with semicolon
- ☐ b. All variables in C++ are identifiers
- ☒ c. ExitExam and exitExam are distinct in C++ ✓
- ☐ d. Every C++ statement should include the header file iostream.h

The correct answer is: ExitExam and exitExam are distinct in C++

QUESTION 4

Correct

Mark 1.00 out of 1.00

Which of the following is true about the statement `a=b;`

- ☐ a. The value of a and b are equal
- ☐ b. The value of b is assigned to variable a and the later change on variable b will affect the value of variable a
- ☒ c. The value of b is assigned to variable a and the later change on variable b will not affect the value of variable a ✓
- ☐ d. The value of variable a is assigned to variable b and the value of variable b is assigned to variable a

The correct answer is: The value of b is assigned to variable a and the later change on variable b will not affect the value of variable a

QUESTION 5

Correct

Mark 1.00 out of 1.00

Which one of the following is true statement about an algorithm?

- ☒ a. An algorithm should take finite amount of time to compute. ✓
- ☐ b. An algorithm may not have a beginning and an end.
- ☐ c. Ambiguous steps are allowed in an algorithm.
- ☐ d. Complex problems cannot have an algorithm.

The correct answer is: An algorithm should take finite amount of time to compute.

QUESTION 6

Incorrect

Mark 0.00 out of 1.00

Which of the following will NOT increase an integer variable "i" by 1?

- ☐ a. i++;
- ☐ b. C. i=i+i;
- ☒ c. D. i=i+1; E. ++i; ❌
- ☐ d. B. i+=1;

The correct answer is: C. i=i+i;

QUESTION 7

Correct

Mark 1.00 out of 1.00

What will happen in the following C++ code snippet?

```
int a = 100, b = 200;
```

```
int *p = &a, *q = &b;
```

```
p = q;
```

- ☐ A. a is assigned to b
- ☐ B. b is assigned to a
- ☒ C. p now points to b ✔️
- ☐ D. q now points to a

Your answer is correct.

The correct answer is:

p now points to b

QUESTION 8

Correct

Mark 1.00 out of 1.00

Given three classes A, B and C, where class C is derived from class B, which is derived from class A. If those classes are public classes and found in different package, then a class C object can access

- ☐ a. Protected, default and public data only in C and B.
- ☒ b. Any data in C and Only Protected and public data in A and B. ✓
- ☐ c. Protected, default and public data only in C.
- ☐ d. Any data in C and only public data in A and B.

The correct answer is: Any data in C and Only Protected and public data in A and B.

QUESTION 9

Correct

Mark 1.00 out of 1.00

If the method is not static; how to invoke in the java programming language?

- ☐ a. Classname.methodname()
- ☒ b. Objectname.methodname() ✓
- ☐ c. Methodname()
- ☐ d. Variablename.methodname

The correct answer is: Objectname.methodname()

QUESTION 10

Correct

Mark 1.00 out of 1.00

Which of the following statement is correct, when an overridden method is called from within a subclass?

- ☐ a. It will always refer to the version of that method defined by the superclass
- ☐ b. Compile time error will occur
- ☒ c. It will always refer to the version of that method defined by the subclass ✓
- ☐ d. The Compiler will choose randomly the method to be executed

The correct answer is: It will always refer to the version of that method defined by the subclass

QUESTION 11

Correct

Mark 1.00 out of 1.00

Which of the following statements not true?

- ☐ a. A public member of a class can be accessed in all the packages
- ☐ b. A protected member of a class can be accessed from its derived class.
- ☐ c. A private member of a class cannot be accessed from its derived class
- ☒ d. A private member of a class cannot be accessed by the methods of the same class ✓

The correct answer is: A private member of a class cannot be accessed by the methods of the same class

QUESTION 12

Incorrect

Mark 0.00 out of 1.00

_____ is one of the basic principles of object oriented programming that use one object in multiple forms.

- ☐ a. Inheritance
- ☒ b. Interface ✗
- ☐ c. Encapsulation
- ☐ d. None of the above

The correct answer is: None of the above

QUESTION 13

Incorrect

Mark 0.00 out of 1.00

Consider the program:

```
public class Student {  
    int id;  
    int section;  
    void display(){  
        System.out.println("ID : " + id);  
        System.out.println("Section : " + section);}  
    public static void main(String[] args){  
        /*Create an object*/  
    }}
```

Which of the following is a correct way to create an object in the main method of the above program?

- ☐ A. Student stud=new Student ();
- ☒ B. Student stud =new Student (3983,2); ❌
- ☐ C. Student stud;
- ☐ D. Student stud = Student (3983,20);

Your answer is incorrect.

The correct answer is:

Student stud=new Student ();

QUESTION 14

Correct

Mark 1.00 out of 1.00

Suppose we want to determine the efficiency of the algorithm, then how we can measure the space factor

- ☒ a. To count the maximum memory required by the algorithm ✔️
- ☐ b. To count the average memory required by the algorithm
- ☐ c. To count the maximum disk space needed by the algorithm
- ☐ d. To count the minimum memory required by the algorithm

The correct answer is: To count the maximum memory required by the algorithm

QUESTION 15

Correct

Mark 1.00 out of 1.00

Two main measures for the efficiency of an algorithm are

- ☐ a. Complexity and capacity
- ☒ b. Time and space ✓
- ☐ c. Processor and memory
- ☐ d. Data and space

The correct answer is: Time and space

QUESTION 16

Incorrect

Mark 0.00 out of 1.00

What is the best time complexity we can achieve to precompute all-pairs shortest paths in a weighted graph?

- ☒ a. $O(n)$ ✗
- ☐ b. $O(n^2)$
- ☐ c. $O(n^3)$
- ☐ d. $O(n^4)$

The correct answer is: $O(n^3)$

QUESTION 17

Correct

Mark 1.00 out of 1.00

What is the time complexity of optimal binary search tree?

- ☐ a. $O(n^2)$
- ☒ b. $O(\log n)$ ✓
- ☐ c. $O(n!)$
- ☐ d. $O(n)$

The correct answer is: $O(\log n)$

QUESTION 18

Incorrect

Mark 0.00 out of 1.00

Which of the following is the space complexity of a recursive algorithm with n levels of recursion and $O(1)$ space complexity in each level?

- ☐ a. $O(1)$
- ☒ b. $O(n)$ ✖
- ☐ c. $O(\log n)$
- ☐ d. $O(n^2)$

The correct answer is: $O(\log n)$

QUESTION 19

Incorrect

Mark 0.00 out of 1.00

Which of the following is the best case time complexity of linear search algorithm?

- ☐ a. $O(1)$
- ☐ b. $O(n^2)$
- ☐ c. $O(\log n)$
- ☒ d. $O(n)$ ✖

The correct answer is: $O(1)$

QUESTION 20

Correct

Mark 1.00 out of 1.00

If the elements '1', '2', '3' and '4' are inserted in a queue, what would be order for the removal?

- ☐ a. 4321
- ☐ b. 3241
- ☒ c. 1234 ✔
- ☐ d. None of the above

The correct answer is: 1234

QUESTION 21

Correct

Mark 1.00 out of 1.00

What is the disadvantage of array data structure?

- ☐ a. Multiple other data structures can be implemented using arrays.
- ☐ b. Elements are stored in contiguous memory blocks.
- ☒ c. The amount of memory to be allocated should be known beforehand. ✓
- ☐ d. Elements of an array can be accessed in constant time.

The correct answer is: The amount of memory to be allocated should be known beforehand.

QUESTION 22

Correct

Mark 1.00 out of 1.00

What is the maximum number of swaps that can be performed in the Selection Sort algorithm?

- ☐ a. 1
- ☒ b. $n-1$ ✓
- ☐ c. $n-2$
- ☐ d. n

The correct answer is: $n-1$

QUESTION 23

Incorrect

Mark 0.00 out of 1.00

What would be the time complexity if user tries to insert the element at the end of the linked list (head pointer is known)?

- ☐ a. $O(\log n)$
- ☐ b. $O(1)$ 🗨️
- ☐ c. $O(n \log n)$
- ☒ d. $O(1)$ ✗

The correct answer is: $O(1)$ 🗨️

QUESTION 24

Incorrect

Mark 0.00 out of 1.00

Which data structure is required to convert the infix to prefix notation?

- ☐ a. Stack
- ☐ b. Queue
- ☐ c. Linked list
- ☒ d. Binary tree ❌

The correct answer is: Stack

QUESTION 25

Correct

Mark 1.00 out of 1.00

Which one of the following is the process of inserting an element in the stack?

- ☐ a. Insert
- ☐ b. Add
- ☒ c. Push ✔️
- ☐ d. None of the above

The correct answer is: Push

QUESTION 26

Correct

Mark 1.00 out of 1.00

Which of the following is a Divide and Conquer algorithm?

- ☐ a. Bubble Sort
- ☒ b. Merge Sort ✔️
- ☐ c. Selection Sort
- ☐ d. Heap Sort

The correct answer is: Merge Sort

QUESTION 27

Correct

Mark 1.00 out of 1.00

-----is a collection of multiple logically related database distributed over a computer network.

- ☐ a. Data Fragmentation
- ☒ b. Distributed Database ✓
- ☐ c. Data Allocation
- ☐ d. Data Replication

The correct answer is: Distributed Database

QUESTION 28

Correct

Mark 1.00 out of 1.00

..... Means that a transaction must execute exactly once completely or not at all.

- ☐ a. Isolation
- ☐ b. Durability
- ☒ c. Atomicity ✓
- ☐ d. Consistency

The correct answer is: Atomicity

QUESTION 29

Correct

Mark 1.00 out of 1.00

A language that used to formulate the schema of database structure is_____.

- ☐ a. Data Control Language
- ☒ b. Data Definition Language ✓
- ☐ c. Data Manipulation language
- ☐ d. Transaction Control Language

The correct answer is: Data Definition Language

QUESTION 30

Correct

Mark 1.00 out of 1.00

A table can have only one

- ☐ a. Foreign key
- ☐ b. Unique Key
- ☐ c. Candidate key
- ☒ d. Primary key ✓

The correct answer is: Primary key

QUESTION 31

Incorrect

Mark 0.00 out of 1.00

Checkpoints are a part of

- ☒ a. Concurrency measures ✗
- ☐ b. Recovery measures
- ☐ c. Authorization measures
- ☐ d. Security measures

The correct answer is: Recovery measures

QUESTION 32

Correct

Mark 1.00 out of 1.00

When the values in one or more attributes being used as a foreign key must exist there must be a corresponding primary key value in another table in the database, we have created_____

- ☐ a. Entity Integrity Constraints
- ☒ b. Referential Integrity Constraint ✓
- ☐ c. Domain Integrity constraint
- ☐ d. Transitive Dependency

The correct answer is: Referential Integrity Constraint

QUESTION 33

Correct

Mark 1.00 out of 1.00

Which of the following is not a feature of DBMS?

- ☐ a. Minimum Duplication and Redundancy of Data
- ☐ b. High Level of Security
- ☐ c. Support ACID Property
- ☒ d. Single-user Access only ✓

The correct answer is: Single-user Access only

QUESTION 34

Correct

Mark 1.00 out of 1.00

Which of the following is not a function of the database?

- ☐ a. Manipulating data
- ☐ b. Managing stored data
- ☐ c. Provide Backup and recovery
- ☒ d. Analyzing code ✓

The correct answer is: Analyzing code

QUESTION 35

Correct

Mark 1.00 out of 1.00

Which of the following is not a recovery technique?

- ☐ a. ARIES Algorithm
- ☐ b. Deferred update
- ☐ c. Immediate update
- ☒ d. Two-phase commit ✓

The correct answer is: Two-phase commit

QUESTION 36

Incorrect

Mark 0.00 out of 1.00

Which of the following is not true about OODBMS (object-oriented database management system).

- ☐ a. The object is persistent
- ☐ b. The key concept is relation
- ☒ c. The key concepts is objects ❌
- ☐ d. Object Identity is unchangeable

The correct answer is: The key concept is relation

QUESTION 37

Correct

Mark 1.00 out of 1.00

Which of the following techniques of query optimization is uses ordering of the operation?

- ☐ a. Cost estimation
- ☒ b. Heuristic- based ✔️
- ☐ c. D. Sematic Query
- ☐ d. Exhaustive search

The correct answer is: Heuristic- based

QUESTION 38

Correct

Mark 1.00 out of 1.00

Which of the following relationship is matched correctly with appropriate example of MANY TO-ONE?

- ☐ a. [Author – Book]
- ☐ b. [Building – Location]
- ☐ c. [Hospital – Patient]
- ☒ d. [Employee – Department] ✔️

The correct answer is: [Employee – Department]

QUESTION 39

Incorrect

Mark 0.00 out of 1.00

During system design system decomposition is influenced by all except_____

- ☒ a. Coupling ❌
- ☐ b. Cohesion
- ☐ c. Users thinking
- ☐ d. System's functionality

The correct answer is: Users thinking

QUESTION 40

Correct

Mark 1.00 out of 1.00

The functional model is represented graphically with ——

- ☐ a. Class diagram
- ☐ b. Object diagram
- ☐ c. State transition diagram
- ☒ d. Use case diagram ✔️

The correct answer is: Use case diagram

QUESTION 41

Correct

Mark 1.00 out of 1.00

Which of the following concept of software engineering is false statement.

- ☐ a. Design focuses on the developer point of view.
- ☒ b. Result of system analysis is a model that includes a subsystem decomposition ✔️
- ☐ c. System Analysis Focuses on the user's view of the system
- ☐ d. Requirements elicitation focus only on the user's view of the system.

The correct answer is: Result of system analysis is a model that includes a subsystem decomposition

QUESTION 42

Incorrect

Mark 0.00 out of 1.00

Which of the following heuristics is not used to identify Entity objects

- ☐ a. Data sources
- ☐ b. Real-world entities
- ☐ c. Forms
- ☒ d. Real-world activities ❌

The correct answer is: Forms

QUESTION 43

Correct

Mark 1.00 out of 1.00

"Consider a system where, a heat sensor detects an intrusion and alerts the security company." What kind of a requirement the system is providing?

- ☐ a. Non-Functional
- ☐ b. Constraints
- ☐ c. Known Requirement
- ☒ d. Functional Requirement ✔️

The correct answer is: Functional Requirement

QUESTION 44

Incorrect

Mark 0.00 out of 1.00

which one of the following is used to Identifying Actors

- ☐ a. Which user groups execute the system's main functions?
- ☒ b. What kind of dependencies are exist among use cases. ❌
- ☐ c. What information does the actor access?
- ☐ d. What are the tasks that the actor wants the system to perform?

The correct answer is: Which user groups execute the system's main functions?

QUESTION 45

Correct

Mark 1.00 out of 1.00

In PHP, which function is used to establish a secure connection to a MySQL database using SSL encryption?

- ☐ a. `secure_connect()`
- ☐ b. `mysql_ssl_connect()`
- ☐ c. `pdo_connect_ssl()`
- ☒ d. `mysqli_ssl_connect()` ✓

The correct answer is: `mysqli_ssl_connect()`

QUESTION 46

Incorrect

Mark 0.00 out of 1.00

Which JavaScript function is used to display a dialog box with an alert message?

- ☐ a. `alert()`
- ☐ b. `confirm()`
- ☐ c. `console.log()`
- ☒ d. `prompt()` ✗

The correct answer is: `alert()`

QUESTION 47

Correct

Mark 1.00 out of 1.00

Which JavaScript method is used to select an HTML element by its ID?

- ☐ a. `getElementsByTag()`
- ☐ b. `getElementsByName()`
- ☐ c. `getElementsByClass()`
- ☒ d. `getElementById()` ✓

The correct answer is: `getElementById()`

QUESTION 48

Correct

Mark 1.00 out of 1.00

Which language is used to add interactivity and dynamic behavior to a web page?

- ☐ a. CSS
- ☒ b. JavaScript ✓
- ☐ c. HTML
- ☐ d. PHP

The correct answer is: JavaScript

QUESTION 49

Correct

Mark 1.00 out of 1.00

Which of the following languages is primarily responsible for defining the structure and content of a web page?

- ☐ a. JavaScript
- ☒ b. HTML ✓
- ☐ c. CSS
- ☐ d. PHP

The correct answer is: HTML

QUESTION 50

Correct

Mark 1.00 out of 1.00

Which PHP function is used to check if a file or directory exists?

- ☐ a. file_exists_check()
- ☐ b. file_check()
- ☐ c. is_file()
- ☒ d. file_exists() ✓

The correct answer is: file_exists()

QUESTION 51

Incorrect

Mark 0.00 out of 1.00

Which PHP function is used to retrieve the number of rows affected by an SQL query?

- ☒ a. `num_rows()` ❌
- ☐ b. `affected_rows()`
- ☐ c. `get_affected_rows()`
- ☐ d. `row_count()`

The correct answer is: `affected_rows()`

QUESTION 52

Correct

Mark 1.00 out of 1.00

Which PHP superglobal array is used to retrieve information about the server and execution environment?

- ☐ a. `$_POST`
- ☒ b. `$_SERVER` ✔️
- ☐ c. `$_SESSION`
- ☐ d. `$_GET`

The correct answer is: `$_SERVER`

QUESTION 53

Correct

Mark 1.00 out of 1.00

Which PHP superglobal array is used to retrieve data sent from an HTML form using the POST method?

- ☒ a. `$_POST` ✔️
- ☐ b. `$_SERVER`
- ☐ c. `$_REQUEST`
- ☐ d. `$_GET`

The correct answer is: `$_POST`

QUESTION 54

Incorrect

Mark 0.00 out of 1.00

An agent type which has condition-action rule and memory to remind the current state of the world and its evolution is called

- ☐ a. Model based reflex
- ☐ b. Goal based
- ☒ c. Utility based ✖
- ☐ d. Simple reflex

The correct answer is: Model based reflex

QUESTION 55

Incorrect

Mark 0.00 out of 1.00

Assume that you are employed as a machine learning expert at the Ministry of Education (MoE). Which machine learning technique you would apply for the task of describing the existing higher education institutions as first generation, second generation and third generation.

- ☐ a. Reinforcement Learning
- ☒ b. Classification ✖
- ☐ c. Regression
- ☐ d. Clustering

The correct answer is: Clustering

QUESTION 56

Incorrect

Mark 0.00 out of 1.00

Consider the definition of AI: "The effort to design and implement intelligent software that solves problems by identifying best action sequence to achieve the desired goal" can be views AI as

- ☐ a. A system that thinks humanly
- ☐ b. A system that acts rationally
- ☒ c. A system that thinks rationally ✖
- ☐ d. A system that acts humanly

The correct answer is: A system that acts rationally

QUESTION 57

Correct

Mark 1.00 out of 1.00

What is completeness of search algorithms?

- ☒ a. Provided that a solution exists, the algorithm will find it ✓
- ☐ b. An estimate of how much information the algorithm needs to store for finding a solution
- ☐ c. An estimate of how many search steps it takes to find a solution
- ☐ d. If there is a difference between the quality of goal states, the agent will find the best one

The correct answer is: Provided that a solution exists, the algorithm will find it

QUESTION 58

Correct

Mark 1.00 out of 1.00

Which one of the view of defining AI concerned with thought processing

- ☐ a. Viewing AI as a system that act omnisciently to its environment
- ☒ b. Viewing AI as a system that think either humanly or think rationally ✓
- ☐ c. Viewing AI as a system that act either humanly or think rationally
- ☐ d. Viewing AI as a system which has complete autonomous

The correct answer is: Viewing AI as a system that think either humanly or think rationally

QUESTION 59

Correct

Mark 1.00 out of 1.00

Which one of the following statement is false

- ☐ a. Validating AI system that think rationally can be made by using its reasoning logic
- ☐ b. Validating AI system that act rationally can be made by its performance
- ☒ c. Validating AI system that thinks humanly is impossible ✓
- ☐ d. Turing test is a mechanism to validate AI system that act humanly

The correct answer is: Validating AI system that thinks humanly is impossible

QUESTION 60

Correct

Mark 1.00 out of 1.00

A context free Grammar is called ambiguous if there exists a string that can have?

- ☐ a. no parse tree
- ☐ b. only one parse tree
- ☒ c. more than one parse tree ✓
- ☐ d. Partial parse tree

The correct answer is: more than one parse tree

QUESTION 61

Incorrect

Mark 0.00 out of 1.00

A context free grammar with production rules: $S \rightarrow aS / \epsilon$ is equivalent to the regular expression

- ☐ a. $a(a+b)^*$
- ☒ b. $a+$ ✗
- ☐ c. a^*
- ☐ d. $(a+b)^*$

The correct answer is: a^*

QUESTION 62

Incorrect

Mark 0.00 out of 1.00

A push down automaton employs _____ data structure.

- ☒ a. Linked List ✗
- ☐ b. Stack
- ☐ c. Queue
- ☐ d. Hash Table

The correct answer is: Stack

QUESTION 63

Correct

Mark 1.00 out of 1.00

Which of the following Machine is Equivalent for Regular grammar?

- ☐ a. Linear bounded automata
- ☐ b. Push down automata
- ☐ c. Turing Machine
- ☒ d. Finite state automata ✓

The correct answer is: Finite state automata

QUESTION 64

Incorrect

Mark 0.00 out of 1.00

Given a context free grammar $G = \{V, \Sigma, P, S\}$ and production $P: a \rightarrow b$. Which one of the following is true?

- ☐ a. $b \in (V + \Sigma)^*$
- ☐ b. $a \in (V + \Sigma)$
- ☒ c. $a \in V$ ✗
- ☐ d. $a \in (V + \Sigma)^*$

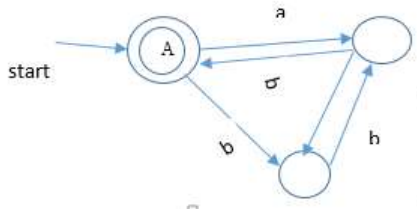
The correct answer is: $b \in (V + \Sigma)^*$

QUESTION 65

Correct

Mark 1.00 out of 1.00

Given a finite automata below, what will be the regular expression that correctly represents the language accepted by it.



- ☐ A. $a^*(a+b)ab(ab+bb+aa^*(a+b)ab)^*$
- ☐ B. $(aa^*(a+b)ab)^*$
- ☒ C. $(a+b)ab(ab+bb+aa^*(a+b)ab)^*$ ✓
- ☐ D. $a^*ab(ab+bb+aa^*(a+b)ab)^*$

Your answer is correct.

The correct answer is:

$(a+b)ab(ab+bb+aa^*(a+b)ab)^*$

QUESTION 66

Incorrect

Mark 0.00 out of 1.00

Identify the technique used to replace run-time computations with compile-time computations?

- ☒ a. Code hoisting ✗
- ☐ b. Invariant computation
- ☐ c. Pee hole optimization
- ☐ d. Constant folding

The correct answer is: Constant folding

QUESTION 67

Incorrect

Mark 0.00 out of 1.00

In which does the compiler translate the source code?

- ☒ a. Machine code ❌
- ☐ b. Both a and b
- ☐ c. Binary code
- ☐ d. Executable code

The correct answer is: Both a and b

QUESTION 68

Incorrect

Mark 0.00 out of 1.00

Through which type of grammar can synthesized attributes can be simulated?

- ☐ a. Ambiguous grammar
- ☐ b. LR grammar
- ☐ c. LL grammar
- ☒ d. None ❌

The correct answer is: LR grammar

QUESTION 69

Correct

Mark 1.00 out of 1.00

What is regular expression for an Identifier?

- ☐ a. Letter(letter_digit)*
- ☐ b. Letter(letter | digit)+
- ☒ c. Letter(letter | digit)* ✔️
- ☐ d. Letter(letter+ | digit*)*

The correct answer is: Letter(letter | digit)*

QUESTION 70

Correct

Mark 1.00 out of 1.00

What is the process of finding a parse tree for a string of tokens called?

- ☐ a. Tokenizing
- ☒ b. Parsing ✓
- ☐ c. Recognizing
- ☐ d. Analyzing

The correct answer is: Parsing

QUESTION 71

Correct

Mark 1.00 out of 1.00

What is the action of parsing the source program into proper syntactic classes?

- ☐ a. Syntax analysis
- ☒ b. Lexical analysis ✓
- ☐ c. Interpretation analysis
- ☐ d. General syntax analysis

The correct answer is: Lexical analysis

QUESTION 72

Incorrect

Mark 0.00 out of 1.00

All are properties of the integrity goal of a computer system, except

- ☐ a. Ensures there was an authorized modification
- ☒ b. Integrity can be compromised by Deception ✗
- ☐ c. Keep resources from active attacks
- ☐ d. Integrity can be compromised by Disclosure

The correct answer is: Integrity can be compromised by Disclosure

QUESTION 73

Correct

Mark 1.00 out of 1.00

Assume, you went to the Ethiopian airport to catch a flight and told the airport personnel your name. For sure, they would ask you some questions like your passport number and your reservation ticket number for some proof of your identity. Then what do we call this process?

- ☒ a. Authentication ✓
- ☐ b. Authorization
- ☐ c. Access control
- ☐ d. All

The correct answer is: Authentication

QUESTION 74

Correct

Mark 1.00 out of 1.00

Assume, you went to the nearby commercial bank's automatic teller machine (ATM) to withdraw money. You inserted your ATM card into the machine, and the machine asked you to enter your personal identification number (PIN). Then, when you type your password, someone intercepts your ATM card number, and which of your security goals is compromised?

- ☐ a. Integrity
- ☐ b. Authorization
- ☒ c. Confidentiality ✓
- ☐ d. Availability

The correct answer is: Confidentiality

QUESTION 75

Correct

Mark 1.00 out of 1.00

Which attack can be used for monitoring children's activity?

- ☐ a. Ransomwares
- ☐ b. Adware
- ☐ c. Worms
- ☒ d. Spyware ✓

The correct answer is: Spyware

QUESTION 76

Correct

Mark 1.00 out of 1.00

_____ is a type of data link layer attack in TCP/IP protocol stack

- ☐ a. ARP spoofing
- ☐ b. DNS spoofing
- ☐ c. SYNC-FLOOD attack
- ☒ d. All except c ✓

The correct answer is: All except c

QUESTION 77

Incorrect

Mark 0.00 out of 1.00

Which one of the following statements is true about worms and Viruses

- ☐ a. Virus requires a host to spread while Worms doesn't require
- ☒ b. Worms requires a user action for replicate while viruses doesn't require ✗
- ☐ c. Worms spreading speed is slower than Virus Spreading speed
- ☐ d. Worms requires a host to spread while viruses doesn't require

The correct answer is: Virus requires a host to spread while Worms doesn't require

QUESTION 78

Incorrect

Mark 0.00 out of 1.00

What is the bit pattern for the first octet of a class-B network address?

- ☐ a. 1110XXXX
- ☐ b. 0XXXXXXX
- ☒ c. 110XXXX ✗
- ☐ d. 10XXXXXX

The correct answer is: 10XXXXXX

QUESTION 79

Correct

Mark 1.00 out of 1.00

Which OSI model layer is responsible for flow control?

- ☐ a. Application Layer
- ☐ b. Session Layer
- ☒ c. Transport Layer ✓
- ☐ d. Presentation Layer

The correct answer is: Transport Layer

QUESTION 80

Correct

Mark 1.00 out of 1.00

Which of the following networking device is used in physical layer?

- ☒ a. Hub ✓
- ☐ b. Router
- ☐ c. Bridge
- ☐ d. Switch

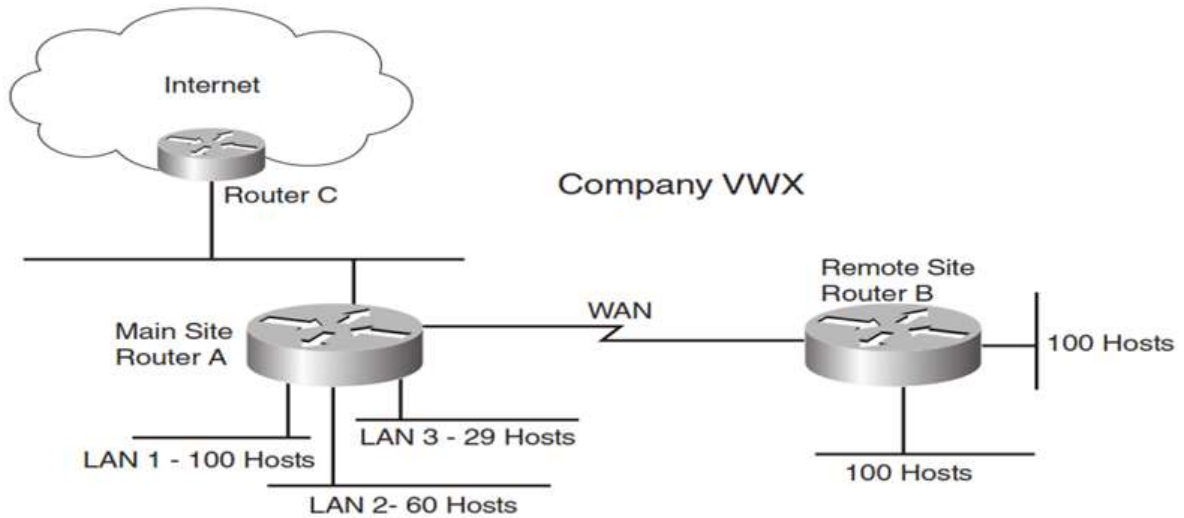
The correct answer is: Hub

QUESTION 81

Incorrect

Mark 0.00 out of 1.00

Assuming that Company VWX has main site and remote site. The main site has three LANs with LAN 1 has 100 hosts, LAN 2 has 60 hosts, and LAN 3 has 29 hosts. The remote site has two LANs, each with 100 hosts. The network uses private addresses. The Internet service provider assigned the company the network 200.100.50.2/26.



The remote site uses the network prefix 192.168.10.0/24. What subnets and masks can you use for the LANs at the remote site and conserve address space?

- ☐ A. 192.168.10.32/28 and 192.168.10.64/28
- ☐ B. 192.168.10.64/26 and 192.168.10.192/26
- ☐ C. 192.168.10.0/25 and 192.168.10.128/25
- ☒ D. 192.168.10.0/30 and 192.168.10.128/30 ❌

Your answer is incorrect.

The correct answer is:

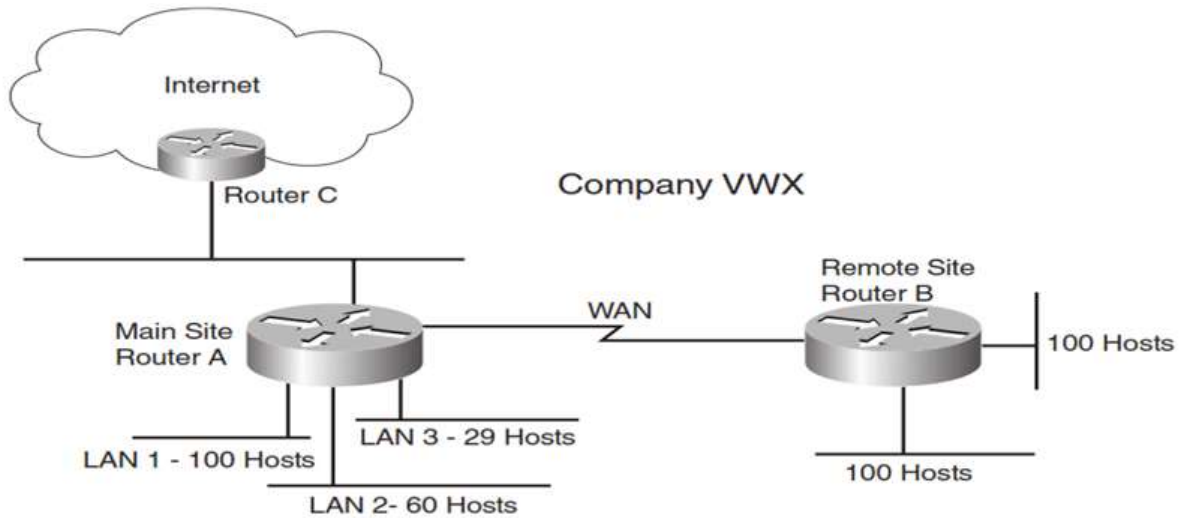
192.168.10.0/25 and 192.168.10.128/25

QUESTION 82

Incorrect

Mark 0.00 out of 1.00

The main site uses the network prefix 192.168.15.0/24. What subnets and masks can you use to provide sufficient addresses for LANs at the main site and conserve address space?



- ☐ A. 192.168.15.0/25 for LAN 1, 192.168.15.128/26 for LAN 2, and 192.168.15.192/27 for LAN 3
- ☐ B. 192.168.15.0/26 for LAN 1, 192.168.15.128/26 for LAN 2, and 172.15.192.0/29 for LAN 3
- ☒ C. 192.168.15.0/27 for LAN 1, 192.168.15.128/26 for LAN 2, and 172.15.192.0/25 for LAN 3 ✖
- ☐ D. 192.168.15.0/100 for LAN 1, 192.168.15.128/60 for LAN 2, and 172.15.192.0/29 for LAN 3

Your answer is incorrect.

The correct answer is:

192.168.15.0/25 for LAN 1, 192.168.15.128/26 for LAN 2, and 192.168.15.192/27 for LAN 3

QUESTION 83

Correct

Mark 1.00 out of 1.00

In DNS concept the process of getting IP address by using its hostname is called: _____

- ☐ a. Binding
- ☒ b. Forward resolution ✓
- ☐ c. Backward resolution
- ☐ d. Resolving

The correct answer is: Forward resolution

QUESTION 84

Incorrect

Mark 0.00 out of 1.00

Let's assume that you know the domain name of Face Book server, but you don't know its IP address. So which network troubleshooting tool you use to get its IP address?

- ☒ a. Nslookup ✗
- ☐ b. Traceroute
- ☐ c. Ping command
- ☐ d. Netstat

The correct answer is: Netstat

QUESTION 85

Incorrect

Mark 0.00 out of 1.00

Where/when Network administrator should have to use the concept of Network Address translation? Network admin should use NAT where: -

- ☐ a. Global IP address is used in the LAN
- ☒ b. He/she don't know type of IP address used in the LAN ✗
- ☐ c. Private IP address is used in the LAN
- ☐ d. Default gateway of LAN isn't configured

The correct answer is: Private IP address is used in the LAN

QUESTION 86

Correct

Mark 1.00 out of 1.00

Where/When we can use DHCP Services? It is recommended to use DHCP when: -

- ☐ a. There are small number of network users
- ☐ b. Network admin went to use same end devices for specific purpose
- ☒ c. Number of devices in the network are very complex ✓
- ☐ d. All are correct

The correct answer is: Number of devices in the network are very complex

QUESTION 87

Correct

Mark 1.00 out of 1.00

Which of the following server type is used to store, process, and respond to the client request for webpages via http/https protocols

- ☐ a. DNS Server
- ☐ b. Http Server
- ☒ c. Web Server ✓
- ☐ d. FTP server

The correct answer is: Web Server

QUESTION 88

Correct

Mark 1.00 out of 1.00

_____ is a network security device that monitors incoming and outgoing network traffic.

- ☐ a. Proxy
- ☐ b. Router
- ☒ c. Firewall ✓
- ☐ d. Gateway

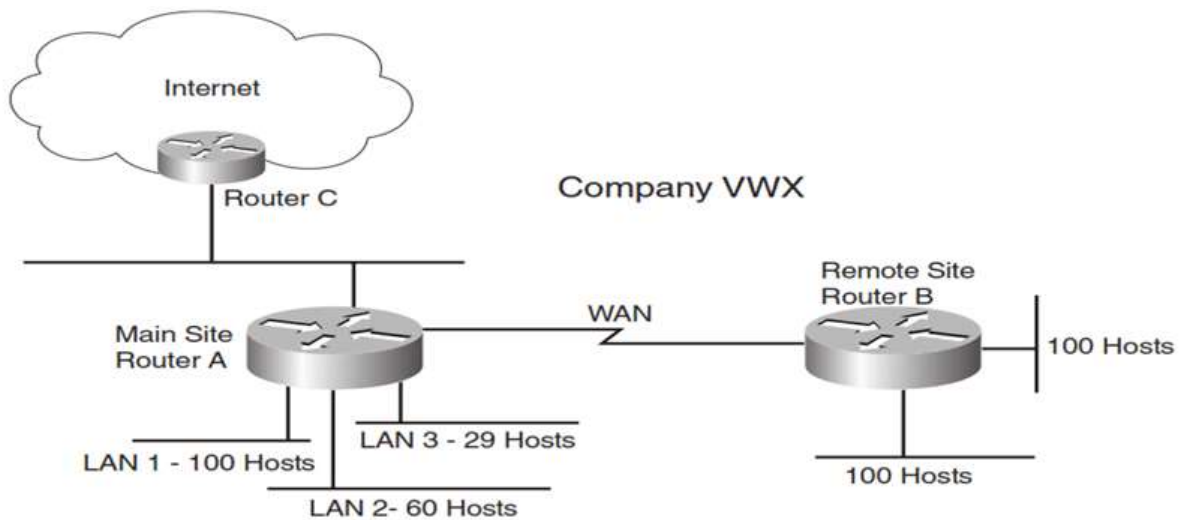
The correct answer is: Firewall

QUESTION 89

Incorrect

Mark 0.00 out of 1.00

Which network and mask would you use for the WAN link to save the most address space?



- ☐ A. 192.168.11.240/29
- ☐ B. 192.168.11.240/30
- ☒ C. 192.168.11.240/27 ❌
- ☐ D. 192.168.11.240/28

Your answer is incorrect.

The correct answer is:

192.168.11.240/28

QUESTION 90

Correct

Mark 1.00 out of 1.00

A process can be terminated due to:

- ☐ a. normal exit
- ☐ b. fatal error
- ☒ c. all of the mentioned ✓
- ☐ d. killed by another process

The correct answer is: all of the mentioned

QUESTION 91

Incorrect

Mark 0.00 out of 1.00

To access the services of operating system, the interface is provided by the:

- ☐ a. assembly instructions
- ☐ b. system calls
- ☒ c. API ❌
- ☐ d. library

The correct answer is: system calls

QUESTION 92

Correct

Mark 1.00 out of 1.00

Which module gives control of the CPU to the process selected by the short-term scheduler?

- ☐ a. none of the mentioned
- ☐ b. Scheduler
- ☐ c. Interrupt
- ☒ d. Dispatcher ✔️

The correct answer is: Dispatcher

QUESTION 93

Incorrect

Mark 0.00 out of 1.00

Which of the following condition is required for deadlock to be possible?

- ☐ a. mutual exclusion
- ☒ b. a process may hold allocated resources while awaiting assignment of other resources ❌
- ☐ c. all of the mentioned
- ☐ d. no resource can be forcibly removed from a process holding it

The correct answer is: all of the mentioned

QUESTION 94

Correct

Mark 1.00 out of 1.00

Which scheduling algorithm allocates the CPU first to the process that requests the CPU first?

- ☐ a. none of the mentioned
- ☐ b. priority scheduling
- ☐ c. shortest job scheduling
- ☒ d. first-come, first-served scheduling ✓

The correct answer is: first-come, first-served scheduling

QUESTION 95

Incorrect

Mark 0.00 out of 1.00

Which one of the following is not true?

- ☐ a. kernel is made of various modules which cannot be loaded in running operating system
- ☐ b. kernel is the first part of operating system to load into memory during booting
- ☒ c. kernel is the program that constitutes the central core of the operating system ✗
- ☐ d. kernel remains in the memory during the entire computer session

The correct answer is: kernel is made of various modules which cannot be loaded in running operating system

QUESTION 96

Correct

Mark 1.00 out of 1.00

A processor performing fetch or decoding of different instruction during the execution of another instruction is called _____

- ☐ a. Super-scaling
- ☐ b. Parallel Computation
- ☐ c. None of the above
- ☒ d. Pipe-lining ✓

The correct answer is: Pipe-lining

QUESTION 97

Incorrect

Mark 0.00 out of 1.00

An instruction that will jump to the next instruction if the condition is met

- ☐ a. Branch instruction
- ☐ b. Jump instruction
- ☒ c. Return instruction ❌
- ☐ d. Skip instruction

The correct answer is: Skip instruction

QUESTION 98

Correct

Mark 1.00 out of 1.00

An interrupt that is caused by register overflow, attempt to divide by zero, an invalid operation code, stack overflow can be?

- ☒ a. Internal interrupt ✔️
- ☐ b. Software interrupt
- ☐ c. None of the above
- ☐ d. External interrupt

The correct answer is: Internal interrupt

QUESTION 99

Incorrect

Mark 0.00 out of 1.00

Computer address bus is

- ☐ a. Multidirectional
- ☒ b. Bidirectional ❌
- ☐ c. Unidirectional
- ☐ d. None of the above

The correct answer is: Unidirectional

QUESTION 100

Incorrect

Mark 0.00 out of 1.00

The simplest way to organize a computer is to have one processor register and an instruction code format with two parts. The first part specifies the operation to be performed and the second specifies .

- ☒ a. operand ❌
- ☐ b. Input
- ☐ c. address
- ☐ d. output

The correct answer is: address

QUESTION 101

Incorrect

Mark 0.00 out of 1.00

Which of the following computer memory is fastest?

- ☐ a. Hard disk
- ☐ b. None of the above
- ☐ c. Register
- ☒ d. RAM ❌

The correct answer is: Register